ORGANIC CHEMISTRY APPLICATIONS

ALKANES

1. Cooking and Heating Fuel

Domestic cooking gas: Methane, propane, and butane are widely used as fuels in gas cylinders and burners. They undergo highly exothermic complete combustion in oxygen to produce a hot, clean blue flame with no soot residue.

2. Motor Engine Fuels

Petrol and diesel for vehicles: Liquid alkanes like octane are used as fuel in cars and generators. They vaporize easily in engine cylinders and burn smoothly under pressure to release large amounts of heat energy that drive the pistons.

3. Lubricants and Protective Greases

Engine oil and vaseline: Heavy liquid and semi-solid alkanes are used to make motor oils and petroleum jelly. Because their non-polar bonds make them chemically inert and insoluble in water, they form a slippery protective barrier that reduces friction and prevents metal rusting.

ALKENES

. Artificial Fruit Ripening (Ethene) Ethene gas is sprayed in agricultural stores to artificially ripen green bananas and tomatoes before sale. As a plant hormone, ethene speeds up fruit enzymes that soften the skin and turn starch into sweet sugar.

2. Manufacturing Plastics (Ethene and Propene)

Ethene and propene are used as raw materials to manufacture plastic bags, bottles, and ropes. Their carbon-carbon double bonds easily break open during addition polymerization to link into long plastic chains.

3. Industrial Ethanol Production (Ethene)

Ethene is reacted with steam under heat and pressure using a phosphoric acid catalyst to make industrial ethanol. Water adds directly across the reactive double bond to yield pure alcohol.

ALKYL HALIDES

1. Inhaled Anaesthetic in Surgery (Halothane)

Halothane is used as an inhaled general anaesthetic to induce unconsciousness in patients during surgical procedures. Its non-polar haloalkane structure enables high solubility in lipid nerve cell membranes, inhibiting nerve impulse transmission safely without the flammability risks of ether.

2. Industrial Dry-Cleaning Solvents (Tetrachloromethane)

Tetrachloromethane and trichloroethene are used as non-aqueous solvents in dry cleaning to extract hydrophobic grease and oil stains from garments. The polar carbon-halogen bonds allow these compounds to dissolve non-polar organic fats without hydrolyzing delicate fabric fibers.

3. Alkylating Reagents in Chemical Synthesis (Chloroethane)

Haloalkanes like chloroethane are used as starting reagents in organic synthesis to manufacture pharmaceuticals, dyes, and Grignard reagents. The electron-deficient carbon atom attached to the halogen undergoes nucleophilic substitution reactions , allowing the transfer of alkyl groups to target nucleophiles.

BENZENE AND METHYLBENZENE

1. Polystyrene Plastic Production (Benzene)

Benzene is converted into ethylbenzene and then styrene monomer to manufacture polystyrene plastics. Its stable aromatic ring remains intact while the side chain undergoes addition polymerization to form durable packaging foams.

2. Dye and Pharmaceutical Synthesis (Benzene)

Benzene is nitrated with concentrated nitric acid and concentrated sulphuric acid to yield nitrobenzene, a key intermediate in aniline dye production. Electrophilic aromatic substitution replaces a ring hydrogen atom with a reactive nitro group .

3. Industrial Organic Solvent (Methylbenzene)

Methylbenzene (toluene) is used as an industrial solvent for dissolving paints, resins, and adhesives. Its non-polar aromatic structure allows it to dissolve lipophilic organic solutes effectively without chemical reaction.

4. Manufacture of Explosives (Methylbenzene)

Methylbenzene is nitrated at elevated temperatures to produce 2,4,6-trinitrotoluene (TNT). The electron-donating methyl group activates the aromatic ring, directing three nitro groups to the 2, 4, and 6 positions.

ALCOHOLS

1. Industrial Solvent (Ethanol and Propan-2-ol)

Ethanol and propan-2-ol (isopropanol) are widely used as industrial solvents in pharmaceuticals, cosmetics, and paints. Their polar hydroxyl group forms hydrogen bonds with water, while their non- polar alkyl chain dissolves organic solutes cleanly.

2. Motor Engine Biofuel (Ethanol)

Ethanol is blended with petrol to make gasohol fuel for internal combustion engines. Its oxygenated hydroxyl group promotes complete combustion in excess air.

3. Antiseptic and Disinfectant (Ethanol - 70%)

Ethanol at 70% concentration is used in hand sanitizers and surgical spirits to sterilize medical equipment and skin. It rapidly denatures microbial proteins and dissolves lipid cell membranes, rendering bacteria and viruses inactive.

4. Polyester Fiber Synthesis (Ethane-1,2-diol)

Ethane-1,2-diol is reacted with benzene-1,4-dicarboxylic acid to manufacture terylene (polyester) synthetic fibers. As a primary diol, its two hydroxyl groups undergo esterification reactions to form long-chain condensation polymers.

PHENOLS

Phenol is used to make antiseptics and disinfectants (TCP and Chloroxylenol) because its hydroxyl group attached directly to the benzene ring denatures bacterial proteins and disrupts microbial cell membranes. 2,4,6-Trichlorophenol is used to make timber preservatives and pesticides because its chlorinated aromatic structure exhibits strong antifungal and antibacterial toxicity against wood-decaying organisms. Phenol is used to make 2-hydroxybenzoic acid (salicylic acid) because its electron-rich aromatic ring readily undergoes electrophilic substitution with carbon dioxide in the Kolbe-Schmitt reaction.

CARBONYL COMPOUNDS

1. Biological Specimen Preservative (Methanal / Formaldehyde)

Methanal gas dissolved in water (formalin) is used to preserve anatomical specimens in laboratories and for embalming. The highly reactive carbonyl carbon cross-links with amino groups in biological proteins, preventing cellular decay and bacterial decomposition.

2. Industrial Solvent for Resins and Plastics (Propanone / Acetone)

Propanone (acetone) is widely used as an industrial solvent for dissolving paints, varnishes, fingernail polish, and cellulose acetate plastics. Its polar carbonyl group interacts favorably with both polar and non-polar organic solutes without reacting with them under ambient conditions.

3. Manufacture of Thermosetting Plastics (Methanal)

Methanal is reacted with phenol or urea to produce Bakelite and urea-formaldehyde resins used in electrical fittings and adhesives. It undergoes condensation reactions where nucleophiles attack the electrophilic carbonyl carbon, linking polymer chains into

a rigid, heat-resistant 3D network lattice.

4. Flavoring Agents and Perfumes (Benzaldehyde and Propan-2-one)

Benzaldehyde and higher ketones are used extensively in food flavorings, cosmetics, and perfumes. Their volatility allows them to evaporate readily, while the polar carbonyl group binds to olfactory receptors to register distinct aromatic aromas like almond and vanilla

CARBOXYLIC ACIDS

1. Food Preservative and Flavoring (Ethanoic Acid / Vinegar)

Dilute ethanoic acid (acetic acid) is used as a food preservative in foods and as a culinary acidifier. Its acidic carboxyl group (donates hydrogen ions to lower the pH of solutions, inhibiting bacterial growth and enzymatic spoilage.

2. Manufacture of Synthetic Polymers and Fibers (Hexanedioic Acid)

Hexanedioic acid (adipic acid) is reacted with hexane-1,6-diamine to manufacture Nylon 6,6 synthetic fibers. Its two terminal carboxylic acid groups undergo condensation polymerization with diamines, forming repeating amide linkages with the elimination of water molecules.

3. Production of Analgesic Medicines (Ethanoic Acid and Salicylic Acid)

Ethanoic acid is reacted with 2-hydroxybenzoic acid (salicylic acid) to synthesize aspirin (acetylsalicylic acid). The carboxyl functional group provides the acyl group necessary to esterify the phenolic hydroxyl group, yielding an active anti-inflammatory pharmaceutical compound.

4. Manufacture of Soaps and Detergents (Long-Chain Fatty Acids)

Long-chain carboxylic acids (fatty acids like stearic acid) are boiled with aqueous sodium hydroxide to produce soap molecules (saponification). The acidic carboxyl group reacts to form an ionic, hydrophilic carboxylate head , while the non-polar hydrocarbon tail dissolves lipophilic oil and dirt

DERIVATIVES OF CARBOXYLIC ACIDS

1. Fragrances, Food Flavorings, and Solvents (Esters)

Esters like ethyl ethanoate and pentyl ethanoate are widely used as synthetic fruit flavorings, perfumes, and solvent thinners. Their polar ester group provides moderate volatility and distinct fruity odors, while allowing them to dissolve organic resins cleanly.

2. Synthesis of Pharmaceuticals and Aspirin (Acid Anhydrides / Ethanoic Anhydride)

Ethanoic anhydride is preferred over ethanoic acid in the industrial synthesis of aspirin from 2- hydroxybenzoic acid. It is a more reactive acylating agent than carboxylic acids, providing a rapid reaction without releasing corrosive hydrogen chloride gas like acyl chlorides do.

3. Reactive Intermediates for Acylation Reactions (Acyl Chlorides / Ethanoyl Chloride)

Ethanoyl chloride is used in organic synthesis laboratories to introduce acetyl groups into alcohols, phenols, and amines to make esters and amides. The strongly electron-withdrawing chlorine atom makes the carbonyl carbon highly electrophilic, allowing fast nucleophilic acyl substitution reactions at room temperature.

4. Synthetic Fiber Production and Structural Plastics (Amides / Polyamides)

Polyamides like Nylon 6,6 and Kevlar are manufactured by reacting dicarboxylic acid derivatives with diamines. The strong polar amide linkages form extensive inter-chain hydrogen bonds, giving these synthetic polymers exceptional tensile strength and high thermal stability.

AMINES

1. Production of Synthetic Dyes (Phenylamine / Aniline)

Phenylamine (aniline) is widely used in the production of synthetic azo dyes for coloring textiles and leather. It is diazotized with nitrous acid below 5 degrees to form benzenediazonium chloride, whose activated aromatic system undergoes coupling reactions with phenols to form brightly colored azo compounds.

2. Synthesis of Pharmaceuticals and Local Anaesthetics (Primary and Secondary Amines)

Primary and secondary amines like diethylamine are essential precursors in the synthesis of pharmaceuticals, including local anaesthetics like lidocaine and antihistamines. The basic nitrogen atom with its lone pair acts as a nucleophile, enabling the creation of basic nitrogenous molecules that bind directly to human cell receptors.

3. Manufacture of Synthetic Polymers (Hexane-1,6-diamine)

Hexane-1,6-diamine is used as a core monomer in the industrial manufacture of Nylon 6,6 synthetic fibers for textiles and ropes. The nucleophilic lone pairs on its two terminal primary amino groups attack carbonyl carbons in dicarboxylic acids to form strong repeating amide linkages.

4. Fabric Softeners and Disinfectants (Quaternary Ammonium Salts)

Quaternary ammonium salts are extensively used as active ingredients in fabric softeners, hair conditioners, and hospital disinfectants. Their positively charged nitrogen head electrostaticly binds to negatively charged fabric surfaces and bacterial cell walls, while the long alkyl tail dissolves greasy residues.

POLYMERS

CONDESATION POYMERS

1. Polyamides (Nylon 6,6 and Nylon 6,10)

Nylon 6,6 is used to make climbing ropes and fishing nets because it possesses high tensile strength and elasticity due to extensive inter-chain hydrogen bonding between its amide linkages. Nylon 6,6 is used to make textile fabrics and carpets because it exhibits exceptional abrasion resistance and durability under repeated physical stress. Nylon 6,6 is used to make industrial ball bearings and gears because it has high impact strength, high mechanical toughness, and a low coefficient of friction. Nylon 6,10 is used to make toothbrush bristles and paintbrush filaments because it has low water absorption and high dimensional stability due to its longer 10-carbon non-polar alkyl chain. Nylon 6,10 is used to make flexible fuel lines and cable sheathing because it possesses high chemical resistance to oils and maintains flexibility without swelling in wet conditions.

2. Polyester (Terylene / PET)

Terylene is used to make crease-resistant clothing and synthetic fabrics because it has rigid polymer chains that resist deformation and maintain shape after washing. PET is used to make shatterproof beverage bottles and food containers because it is lightweight, high-density, and impermeable to gases like carbon dioxide. Terylene is used to make industrial tire cords and conveyor belts because it has high dimensional stability, high tensile strength, and minimal thermal stretching under heavy loads.

ADDITIONAL POLYMERS

1. Polythene (Polyethene)

Low-Density Polythene (LDPE) is used to make plastic squeeze bottles and film bags because it has high flexibility and low density due to its branched polymer chains. High-Density Polythene (HDPE) is used to make rigid milk crates and water pipes because it has high tensile strength and chemical resistance due to its closely packed, linear polymer chains.

2. Polypropene (Polypropylene)

Polypropene is used to make heavy-duty ropes and packaging straps because it possesses high tensile strength and high fatigue resistance under repeated bending. Polypropene is used to make microwaveable food containers because it has a high melting point and superior chemical inertness against hot organic oils.

3. Polyvinyl Chloride (PVC / Polychloroethene)

Unplasticized PVC is used to make rigid water pipes and window frames because it has high structural rigidity, high density, and excellent weather resistance. Plasticized PVC is used to make flexible electrical cable insulation and raincoats because the added plasticizers decrease intermolecular forces to grant high flexibility and electrical resistance.

4. Polystyrene (Polyphenylethene)

Expanded Polystyrene is used to make protective packaging for electronics and disposable coffee cups because it has low thermal conductivity and exceptional shock-absorption properties due to its trapped air pockets. Rigid Polystyrene is used to make clear plastic cutlery and CD cases because it possesses high transparency, high rigidity, and easy mouldability.

5. Perspex

Perspex is used to make aircraft windshields and aquarium display panels because it has exceptionally high optical clarity and high impact resistance compared to standard glass. Perspex is used to make outdoor illuminated signboard displays because it exhibits outstanding weather resistance and high UV degradation stability.

6. Polytetrafluoroethene (PTFE / Teflon)

PTFE is used to make non-stick coatings for cookware and frying pans because it has an extremely low coefficient of friction and high thermal stability due to strong carbon-fluorine bonds. PTFE is used to make chemical-resistant gaskets and seals because it displays total chemical inertness to corrosive acids, alkalis, and organic solvents.

7. Natural Rubber (Polyisoprene)

Unvulcanized Natural Rubber is used to make latex surgical gloves and adhesives because it possesses high elasticity and flexibility due to the coiling of its cis-1,4-polyisoprene chains. Vulcanized Natural Rubber is used to make heavy-vehicle vehicle tires because heating it with sulfur forms disulfide cross-links between chains, granting high tensile strength, high elasticity, and thermal durability.

8. Synthetic Rubber

SBR is used to make passenger car tire treads because it has superior abrasion resistance and higher heat tolerance than natural rubber under friction. SBR is used to make shoe soles and industrial conveyor belting because it possesses high resistance to mechanical wear and high resistance to swelling in polar fluids.

9. Neoprene

Neoprene is used to make wet suits for diving because it has low thermal conductivity and high flexibility due to its cellular gas-filled rubber matrix. Neoprene is used to make industrial oil hoses and engine gaskets because it exhibits high resistance to degradation

by oils, petroleum fuels, and atmospheric ozone due to its chlorine- stabilized structure.

THERMOSETTING POLYMERS

1. Phenol-Formaldehyde Resin (Bakelite)

Bakelite is used to make electrical switches, plugs, and socket casings because it is a non- conducting material with a rigid 3D cross-linked network that prevents electrical flow. Bakelite is used to make heat-resistant cookware handles and saucepan knobs because it is a thermosetting polymer that does not melt, soften, or conduct heat at elevated temperatures.

2. Melamine-Formaldehyde Resin (Melamine)

Melamine is used to make shatterproof dinnerware, plates, and cups because it has high mechanical hardness, high scratch resistance, and exceptional durability under physical impact. Melamine is used to make decorative heat-resistant countertop laminates (Formica) because its dense 3D cross-linked structure grants total resistance to heat, water absorption, and chemical staining.

3 Urea-Formaldehyde Resin

Urea-Formaldehyde is used to make electrical fittings and light switches because it has high electrical insulation properties and can be produced in light, translucent colors unlike Bakelite. Urea-Formaldehyde is used to make wood adhesives and particleboard binders (MDF) because its 3D cross-linked structure forms strong, heat-resistant adhesive bonds with cellulose fibers.

4 Epoxy Resins (Polyurethanes / Araldite)

Epoxy resins are used to make heavy-duty structural adhesives (Araldite) because they form rigid 3D cross-linked networks that bond strongly to metals, glass, and ceramics with minimal shrinkage upon curing. Epoxy resins are used to make protective surface coatings for circuit boards and industrial floors because they exhibit exceptional chemical inertness, high water resistance, and high abrasion toughness.